

Supplementary Material for “Topological Segmentation Error and Boundary-Condition Tuning in Computed Coronary FFR: A Controlled In Silico Study”

Fadillah Yamin, Mohd Azan Mohammed Sapardi, Ming Kwang Tan, Xin Wang, and Mohd-Zulhilmi Ismadi

S1. ANALYSIS PIPELINE

Fig. S1 shows the analysis pipeline. The reduced-order arm runs end to end without manual intervention: every instance, error type, protocol and bed is solved in one automated run from the cohort list, and every table and figure is generated from its output. The 3D case study is a separate branch. A case package (centerline, outlet boundary conditions and probe positions) is exported to the CFD machine, which returns the solved FFR and the as-meshed radius; the radius is used to rebuild the reduced-order counterpart.

Cohort selection (step 3) shuffles the 6944 instances with a fixed seed (20260918) and fills the six bands of leaky-bed baseline FFR scarcest first, rotating the host vessel type and taking at most two instances per tree and one per host vessel and band. Each host vessel keeps only the lesion positions and lengths that meet the eligibility criteria, so it contributes fewer than 32 instances when a criterion fails at one position. The reduced-order solve (step 7) starts from Poiseuille flow, averages successive flow iterates with a relaxation factor of 0.5, and stops when the largest flow change falls below 10^{-8} of the largest flow or after 200 iterations; a solve that reaches the cap is recorded as not converged, and none did.

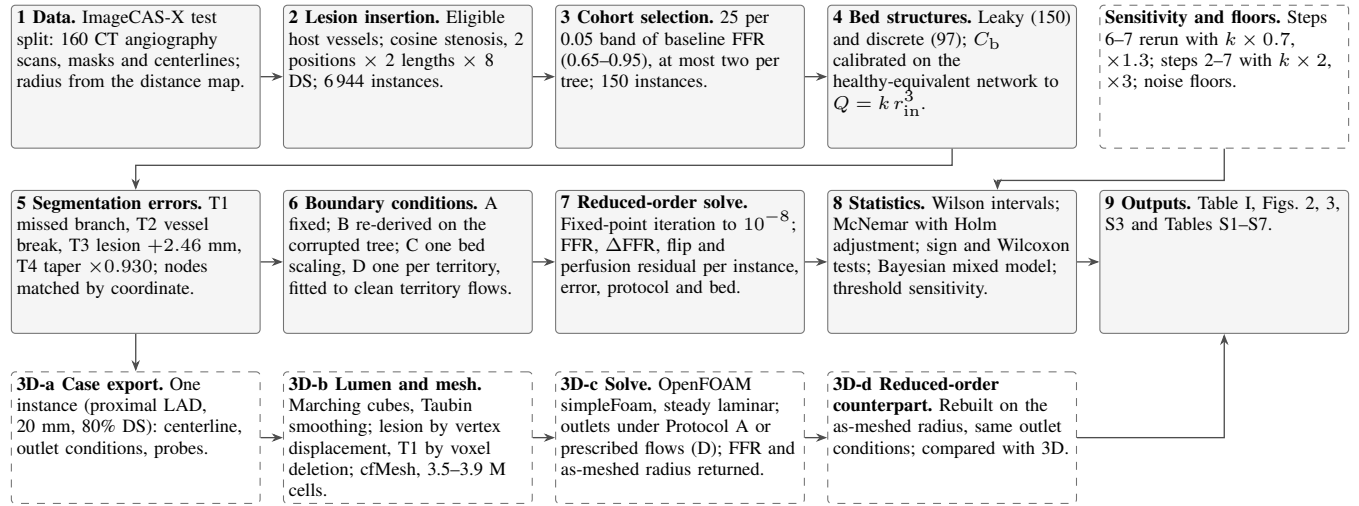


Fig. S1. Analysis pipeline. Shaded boxes: reduced-order arm, run for every instance. Dashed boxes: sensitivity and noise-floor analyses (top right) and the 3D case study (bottom row), whose steps 3D-b and 3D-c ran on the CFD machine.

Fig. S2 shows the inserted stenosis on one host vessel. The narrowing follows (1) of the main text: it is set by the taper-fit radius r_{fit} , not by the local image radius r , and the lumen is never widened.

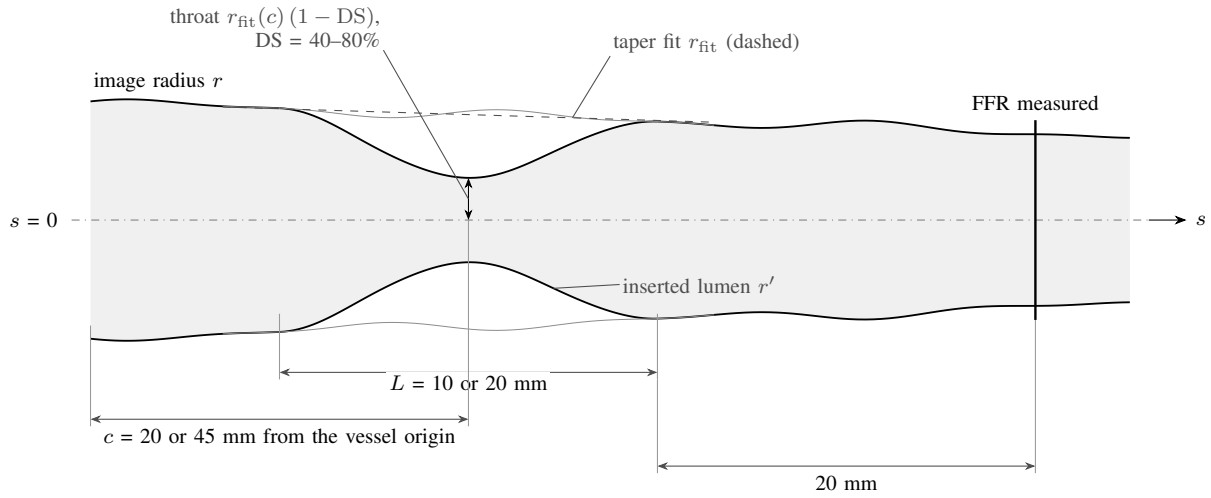


Fig. S2. Inserted stenosis, drawn at 60% DS with $c = 20$ mm and $L = 20$ mm (axial scale compressed). The cosine narrowing is axisymmetric about the centerline and scaled from the taper-fit radius r_{fit} ; outside the lesion and wherever the image radius is already smaller, the lumen is unchanged. Gray line: image radius before insertion. FFR is read 20 mm beyond the distal edge of the lesion.

S2. EXCLUSIONS

Of 6944 swept instances, 150 were selected and 97 from the discrete arm; 2802 corrupted models were defined (an applicable error type under each of Protocols A–C) and 2599 solved, all of which converged. The missed branch was applicable to 77 of 97 (discrete) and 118 of 150 (leaky) instances; the others had no side branch distal to the lesion that could be deleted. The vessel break was not applicable to one instance in each bed, in which the 25 mm truncation point fell at the start of a centerline segment. Protocols C and D are defined only when at least two perfusion territories retain a vessel, so Protocol D was solved on the same instances as Protocol C. A Protocol C fit that ends at a search bound is recorded as a failure; none did in the ablation or in the demand runs (Section S5-B), and 10 of 4940 noise-floor draws did (Section S6). Protocol D scales each territory within $10^{\pm 3}$; the scaling reached 10^3 in two discrete-bed missed-branch models of one tree, which kept perfusion residuals of 0.08 and 0.11 and $|\Delta\text{FFR}| < 0.05$ and are retained. All 769 Protocol D solves converged. Under Protocol A a vessel break can remove every node that carries bed outflow (36 instances in the discrete bed, 2 in the leaky bed). Table S1 lists every exclusion; all other cells were solved for every applicable instance.

TABLE S1
CELLS WITH EXCLUDED MODELS

Bed	Error	Protocol	Solved	Excluded (reason: count)
Discrete	T1	all	77 of 97	no side branch distal to the lesion: 20
Discrete	T1	C	44	fewer than two territories with surviving vessels: 33
Discrete	T2	all	96 of 97	truncation point at a segment start: 1
Discrete	T2	A	60	no node with bed outflow remains: 36
Discrete	T2	C	60	fewer than two territories with surviving vessels: 36
Leaky	T1	all	118 of 150	no side branch distal to the lesion: 32
Leaky	T1	C	71	fewer than two territories with surviving vessels: 47
Leaky	T2	all	149 of 150	truncation point at a segment start: 1
Leaky	T2	A	147	no node with bed outflow remains: 2
Leaky	T2	C	100	fewer than two territories with surviving vessels: 49

S3. SIDE-BRANCH LOSS

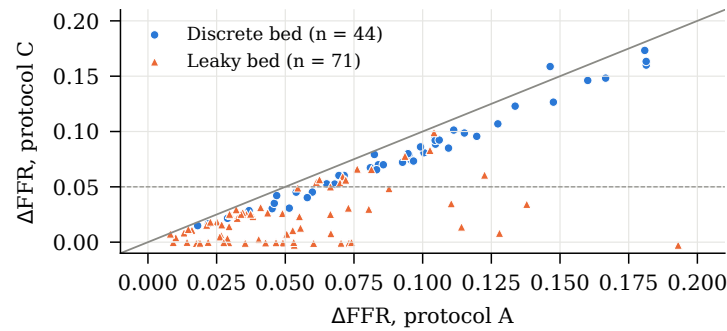


Fig. S3. Change in FFR caused by a missed side branch under fixed boundary conditions (Protocol A) and after tuning to territory flows with one global scaling (Protocol C), paired by instance. The solid line is the identity and the dashed line marks $\Delta\text{FFR} = 0.05$. Under per-territory tuning (Protocol D) the median shift was 0.000 (discrete) and 0.004 (leaky).

S4. MIXED-EFFECTS MODEL OF FLIPS

TABLE S2
BAYESIAN MIXED-EFFECTS LOGISTIC MODEL OF FLIPS: POSTERIOR MEAN LOG-ODDS (95% INTERVAL)

Term	Discrete bed	Leaky bed
Intercept	-2.63 (-2.85, -2.42)	-2.39 (-2.61, -2.18)
T2	0.69 (0.31, 1.07)	1.81 (1.47, 2.15)
T3	-1.92 (-2.40, -1.44)	-2.49 (-3.12, -1.87)
T4	-1.16 (-1.59, -0.73)	-1.00 (-1.45, -0.54)
B	-1.54 (-1.92, -1.17)	-1.76 (-2.27, -1.26)
C	-0.69 (-1.10, -0.28)	-0.99 (-1.42, -0.57)
T2 $\times$ B	0.06 (-0.55, 0.66)	-1.89 (-2.66, -1.12)
T3 $\times$ B	1.26 (0.41, 2.10)	1.13 (-0.06, 2.32)
T4 $\times$ B	0.72 (-0.08, 1.51)	-1.26 (-2.87, 0.35)
T2 $\times$ C	-0.73 (-1.46, 0.00)	-2.00 (-2.76, -1.24)
T3 $\times$ C	0.44 (-0.39, 1.28)	0.78 (-0.25, 1.81)
T4 $\times$ C	-0.10 (-0.89, 0.68)	0.21 (-0.54, 0.96)
Severity (% DS)	0.019 (0.016, 0.022)	0.019 (0.016, 0.022)
Length (mm)	-0.04 (-0.05, -0.02)	-0.08 (-0.09, -0.06)

Reference levels: T1 (missed branch) and Protocol A. Baseline-band terms are omitted from the table. Random intercepts for patient and lesion slot. Fitted by variational inference; intervals are posterior mean ± 1.96 posterior standard deviations from the variational approximation.

At the reference levels the model agrees with the paired tests: under fixed boundary conditions a vessel break carries higher odds of a flip than a missed branch and the caliber errors lower odds, and for the missed branch Protocols B and C carry lower odds than Protocol A, in both beds. In the leaky bed the T2 $\times$ B and T2 $\times$ C interactions offset the T2 main effect, so under re-derived or tuned boundary conditions the two topological errors have similar odds, as in Table I of the main text.

S5. SENSITIVITY ANALYSES

A. Perfusion-Check Threshold

Table S3 repeats the proportion of models that pass the perfusion check while materially wrong at the primary threshold of 10% and at the 13% and 16% sensitivities, for the topological errors and for all error types.

TABLE S3
PROPORTION OF MODELS THAT PASS THE PERFUSION CHECK AND ARE MATERIALLY WRONG, % (WILSON 95% INTERVAL), BY
PERFUSION-CHECK THRESHOLD

Bed	Errors	Protocol	<i>n</i>	10%	13%	16%
Discrete	T1+T2	A	137	12 (8–19)	23 (17–31)	31 (24–40)
Discrete	T1+T2	B	137	1 (0–5)	11 (7–17)	20 (14–27)
Discrete	T1+T2	C	104	19 (13–28)	28 (20–37)	38 (30–48)
Discrete	All	A	331	7 (5–10)	11 (8–15)	15 (11–19)
Discrete	All	B	331	1 (0–2)	5 (3–7)	8 (6–12)
Discrete	All	C	298	10 (7–14)	13 (9–17)	16 (13–21)
Leaky	T1+T2	A	218	20 (15–26)	30 (25–37)	38 (31–44)
Leaky	T1+T2	B	218	5 (3–8)	5 (3–9)	6 (3–9)
Leaky	T1+T2	C	171	8 (5–13)	9 (6–15)	9 (6–15)
Leaky	All	A	518	10 (8–13)	15 (12–18)	18 (15–21)
Leaky	All	B	518	2 (1–4)	2 (1–4)	2 (1–4)
Leaky	All	C	471	11 (8–14)	11 (9–14)	11 (9–14)

Materially wrong: $|\Delta\text{FFR}| > 0.05$. The 13% and 16% thresholds were analyzed as sensitivities. *n* counts models with a defined perfusion residual: T2 under Protocols A and B in the leaky bed, *n* = 100; T2 under Protocol B in the discrete bed, *n* = 60.

B. Hyperemic Demand

The demand constant *k* was scaled by 0.7 and 1.3 and the full ablation was repeated with the same cohort (Table S4). The ordering of topological against caliber errors under fixed boundary conditions, and of Protocols B and C against Protocol A for the topological errors, was unchanged at both levels in both beds. The taper, which flipped fewer decisions under Protocol C than under Protocol A at the primary demand, flipped one and two more at 1.3 times the demand (7 against 6 of 97; 11 against 9 of 150). The magnitudes changed with demand, which moves the clean FFR of every instance (leaky-bed median 0.869, 0.801 and 0.741 at 0.7, 1.0 and 1.3 times the demand) and therefore the number of instances near the threshold, as well as the pressure loss across the lesion. The proportion of tuned topological-error models that pass the perfusion check while materially wrong fell to 3% in the leaky bed at 0.7 times the demand, comparable with the simulated floor for correct anatomy (3.9% at the primary demand); in the discrete bed it remained between 12% and 25%.

TABLE S4
SENSITIVITY TO HYPEREMIC DEMAND

Bed	Demand scale	Flip, % (Protocol A)		Topological flip, %		Passes and wrong, % (C, topological)	Missed-branch ΔFFR median, A / C
		Topological	Caliber	B	C		
Discrete	0.7	32 (25–40)	6 (3–10)	12 (8–18)	18 (12–27)	12 (7–19)	0.070 / 0.057
	1.0	33 (26–41)	10 (7–15)	14 (10–20)	19 (13–28)	19 (13–28)	0.095 / 0.077
	1.3	40 (32–48)	5 (2–9)	16 (11–22)	19 (13–28)	25 (18–34)	0.106 / 0.091
Leaky	0.7	20 (16–26)	6 (4–9)	7 (4–10)	9 (5–14)	3 (1–7)	0.032 / 0.010
	1.0	32 (26–38)	6 (4–9)	5 (3–8)	8 (5–13)	8 (5–13)	0.047 / 0.015
	1.3	37 (32–43)	4 (2–7)	5 (3–9)	8 (4–13)	12 (8–18)	0.057 / 0.021

Demand scale 1.0 is the primary analysis (*k* = 562 s⁻¹). Percentages with Wilson 95% intervals in parentheses. Missed-branch ΔFFR is over the instances in which Protocol C was defined.

C. Demand Replication With Re-Selected Cohorts

Because hyperemic demand follows the segmented inlet radius, the modeled flow was below thermodilution values for normal-sized arteries. The whole pipeline (stenosis sweep, stratified selection with the same rule and seed, discrete-arm eligibility, Protocols A–D) was therefore repeated with *k* doubled and tripled, so that the reselected cohorts again span 0.80 at the higher flow. At three times the demand only 17 instances met the discrete-bed eligibility criteria, mostly because the healthy-equivalent network lost more than 10% of the aortic pressure, so the discrete arm is reported at twice the demand only (Table S5).

TABLE S5
DEMAND REPLICATION: FLIP AND PASSES-AND-WRONG RATES, % (WILSON 95% INTERVAL)

Bed	k scale	Inst.	Flip, A		Topological flip		Topological passes and wrong			Caliber passes and wrong		
			Topol.	Caliber	C	D	B	C	D	B	C	D
Discrete	1	97	33 (26–41)	10 (7–15)	19 (13–28)	8 (4–14)	1 (0–4)	19 (13–28)	20 (14–29)	0 (0–2)	5 (2–9)	18 (13–24)
	2	47	50 (38–62)	1 (0–6)	20 (11–32)	8 (3–18)	5 (2–12)	14 (7–26)	20 (11–32)	0 (0–4)	1 (0–6)	14 (8–22)
Leaky	1	150	32 (26–38)	6 (4–9)	8 (5–13)	6 (3–10)	4 (2–7)	8 (5–13)	5 (3–10)	0 (0–1)	12 (9–17)	4 (2–7)
	2	150	34 (29–40)	5 (3–8)	7 (4–11)	3 (1–7)	2 (1–4)	5 (3–10)	3 (1–7)	0 (0–1)	13 (10–18)	5 (3–8)
	3	150	33 (28–39)	5 (3–8)	6 (4–11)	6 (4–11)	4 (2–7)	7 (4–12)	4 (2–8)	0 (0–2)	12 (9–16)	3 (1–5)

Scale 1 is the primary analysis ($k = 562 \text{ s}^{-1}$, primary cohort). Scales 2 and 3 use cohorts reselected at that demand. Passes and wrong: perfusion residual below 10% and $|\Delta\text{FFR}| > 0.05$; denominators include models without a defined residual, which do not pass.

D. Grey Zone

A change of classification at 0.80 between two values close to the threshold falls inside the diagnostic grey zone of 0.75–0.85. Table S6 separates the flips that end beyond this zone, with the corrupted FFR outside 0.75–0.85 on the other side of 0.80 from the clean FFR. Of the clean models, 30% (discrete) and 33% (leaky) lay inside the zone.

TABLE S6
FLIPS AND FLIPS BEYOND THE GREY ZONE (0.75–0.85), % OF MODELS (WILSON 95% INTERVAL)

Bed	Protocol	Topological errors			Caliber errors		
		n	Flip	Beyond zone	n	Flip	Beyond zone
Discrete	A	137	33 (26–41)	20 (15–28)	194	10 (7–15)	0 (0–2)
	B	173	14 (10–20)	5 (2–9)	194	8 (5–12)	0 (0–2)
	C	104	19 (13–28)	9 (5–16)	194	8 (5–12)	1 (0–3)
	D	104	8 (4–14)	2 (1–7)	194	11 (7–16)	1 (0–3)
Leaky	A	265	32 (26–38)	23 (18–28)	300	6 (4–9)	0 (0–1)
	B	267	5 (3–8)	0 (0–1)	300	1 (1–3)	0 (0–1)
	C	171	8 (5–13)	1 (0–3)	300	4 (2–7)	1 (0–2)
	D	171	6 (3–10)	0 (0–2)	300	4 (2–6)	0 (0–1)

S6. NOISE FLOORS

The repeat-measurement floor uses the published standard deviation of the test–retest difference in invasive FFR, 0.018. The simulated floor uses the clean anatomy of every instance, with 20 draws per instance. Each draw perturbs cardiac output (relative within-subject standard deviation 0.05), mean aortic pressure (0.056), blood viscosity through hematocrit (0.02), the share of flow to each territory (0.10) and the measured territory flows (within-subject coefficient of variation 0.083), and the model is then tuned to the perturbed flows as in Protocol C. Heart rate is not represented in the steady model. The territory-share and measurement terms lie at the low end of published ranges, so the simulated floor may understate physiological variation. Table S7 gives the result. The repeat-measurement floor is 5.0–6.5% across cells and the simulated floor 6.2% and 7.2%.

TABLE S7
SIMULATED FLOOR: CORRECT ANATOMY TUNED TO NOISY PERFUSION

Bed	Instances	Draws solved	Flip, %	Median $ \Delta\text{FFR} $	95th pct. $ \Delta\text{FFR} $	Pass, %	Pass and wrong, %
Discrete	97	1940	6.2 (5.2–7.3)	0.012	0.050	72	3.1 (2.4–4.0)
Leaky	150	2990	7.2 (6.3–8.2)	0.012	0.053	72	3.9 (3.2–4.6)

S7. THREE-DIMENSIONAL CASE STUDY: MESH CHECKS

The mask-to-solution path was scripted with no manual repair, and the lesion-free, baseline and missed-branch solves converged. Three acceptance rules were met with deviations. (i) The throat-size gate compared the ratio of the as-built throat radius to the undeformed meshed radius with the intended radial scale (within 1%), rather than the absolute throat radius with the reduced-order target, because the two pipelines define the radius differently. (ii) A surface self-intersection flag arises from the segmentation mask, not from the mesher. (iii) The strict mesh-quality check flags poorly shaped cells (face-tetrahedra decompositions) 1.43 mm from the throat center, inside the planned 2 mm exclusion distance; the lesion-free mesh carries a similar count, and the standard quality check passes. To test whether these cells affect the result, the baseline was re-meshed with a finer throat zone (Table S8). The acceptance criterion, a change smaller than the discretization uncertainty of 5.5×10^{-4} ,

was not met: the finer zone raised the measurement-point FFR by 0.0019 and 0.0021, while a regenerated mesh at the production resolution reproduced the original value. The second variant missed the strict residual limit at the iteration budget (final velocity residual 1.08×10^{-5} against 10^{-5}) with its inlet pressure steady to 4×10^{-4} Pa over the last 500 iterations; its value is reported as returned. This is a resolution sensitivity of about 0.2% of the FFR, against an error effect of 0.074 in the same case.

All 3D-to-0D comparisons use a reduced-order counterpart rebuilt on the meshed radius, which is wider than the reduced-order radius (main text, Section III-D).

TABLE S8

THROAT-ZONE RESOLUTION TEST, BASELINE LUMEN. THE 8.31 M-CELL MESH DID NOT MEET THE STRICT RESIDUAL LIMIT AT 3 000 ITERATIONS (SEE TEXT).

Mesh	Cells (M)	FFR	Δ FFR	Outlet flow change, %	Flagged cell to throat, mm
Control: 25 μ m, regenerated	3.66	0.8698	+0.0000	0.00	1.43
12.5 μ m, ± 4 mm	6.24	0.8716	+0.0019	0.24	0.70
12.5 μ m, interfaces +1 mm	8.31	0.8718	+0.0021	0.25	0.70